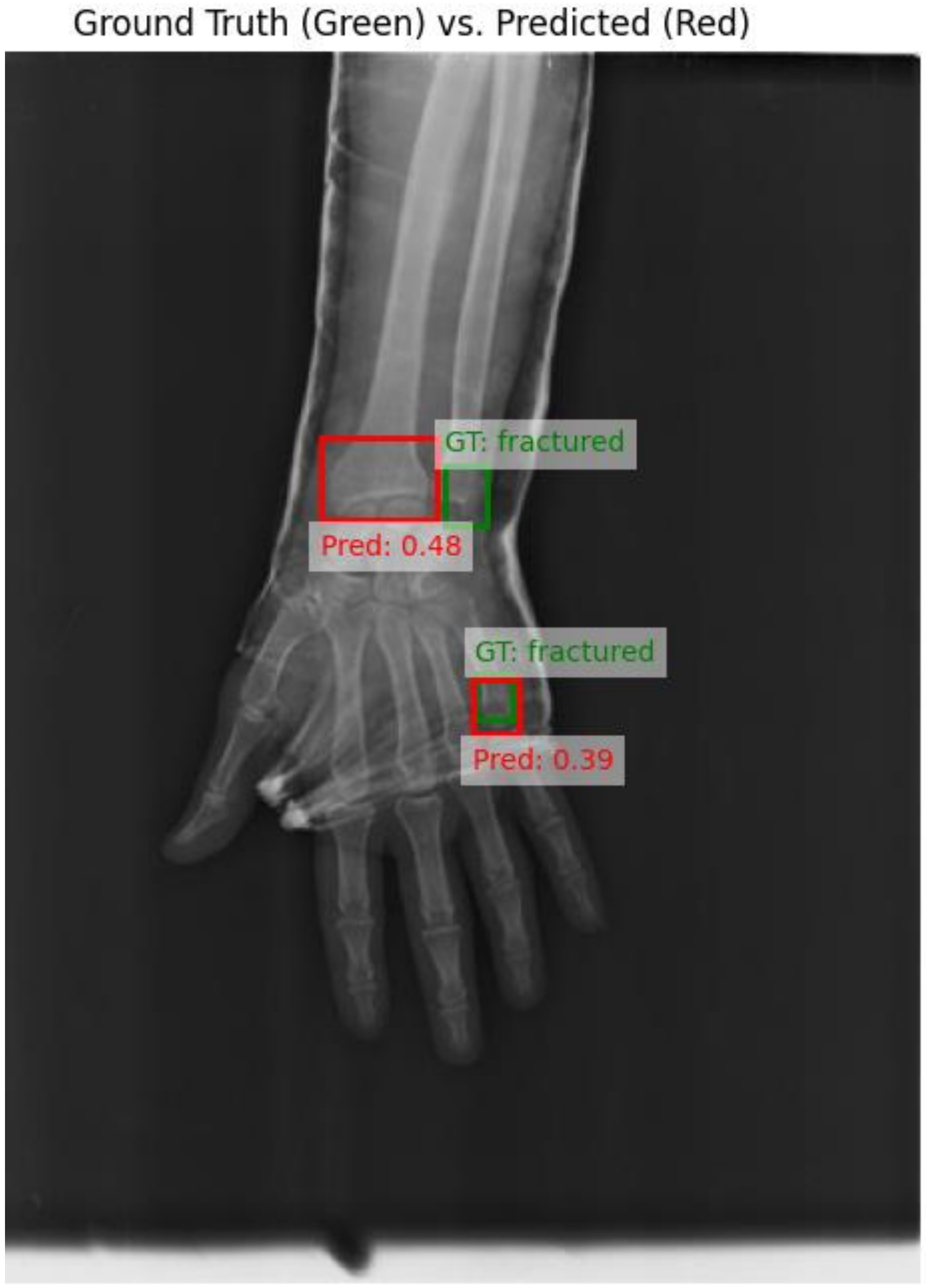
Bone fracture detection from X-rays



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Computer Vision – MSc IDS HS24

Hochschule Luzern – March 12, 2025

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Introduction

- **Why**: Prompt detection prevents complications and aids proper treatment
- **Challenges**: Subtle fractures, overlapping structures, and support for radiologists
- **AI Approach**: Object detection & segmentation to automate fracture identification
- **Overall goal**: Improve diagnostic accuracy and efficiency in clinical workflows

Aim of the current project: to explore the performance of object detection and segmentation approaches, and compare them to traditional image classification. Give a segmentation tool to further develop future models.

Dataset – FracAtlas[1]

- 4083 X-ray images of bones
- annotations indicating the presence of fractures

# of images	3366	61	574	82


not fractured

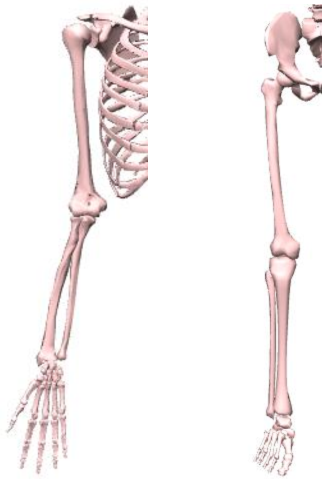

test

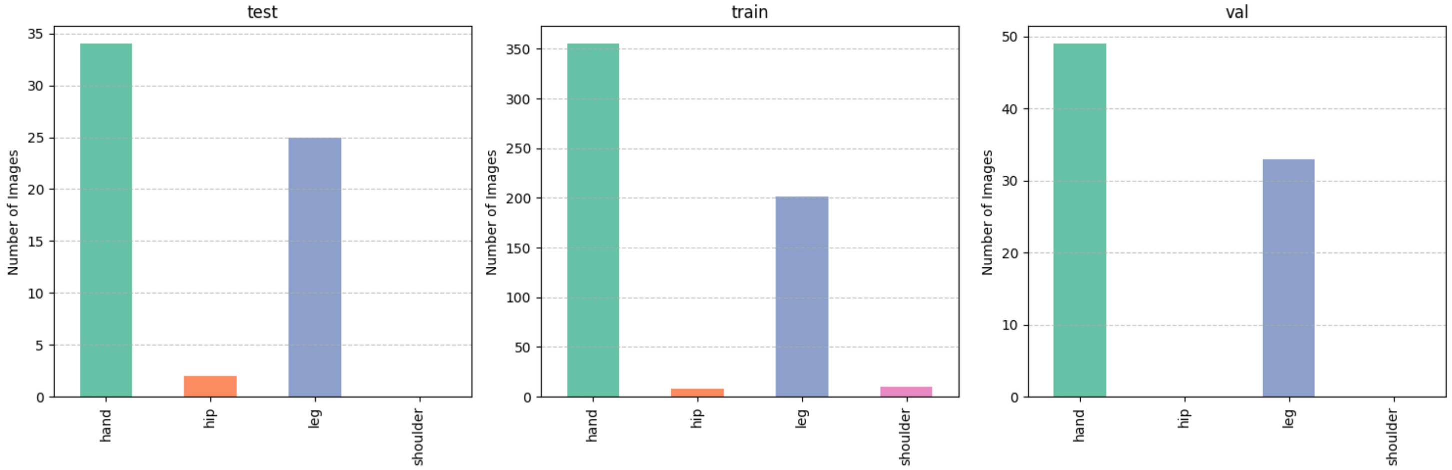

train

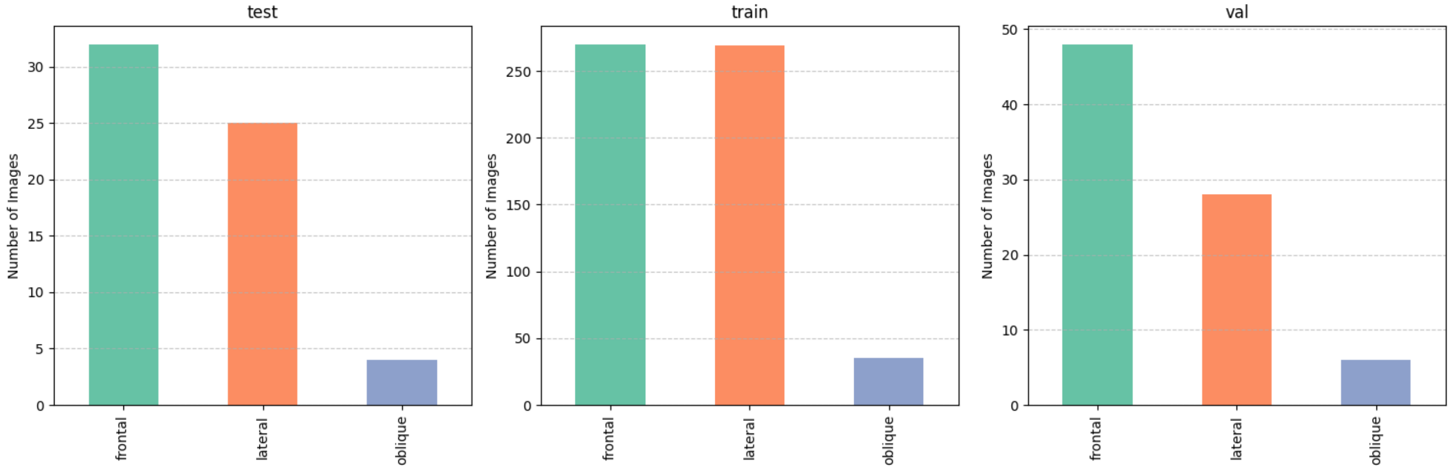

val

[1] Abedeen *et al.* (2023) FracAtlas: A Dataset for Fracture Classification, Localization and Segmentation of Musculoskeletal Radiographs. *Sci Data* **10**, 521.

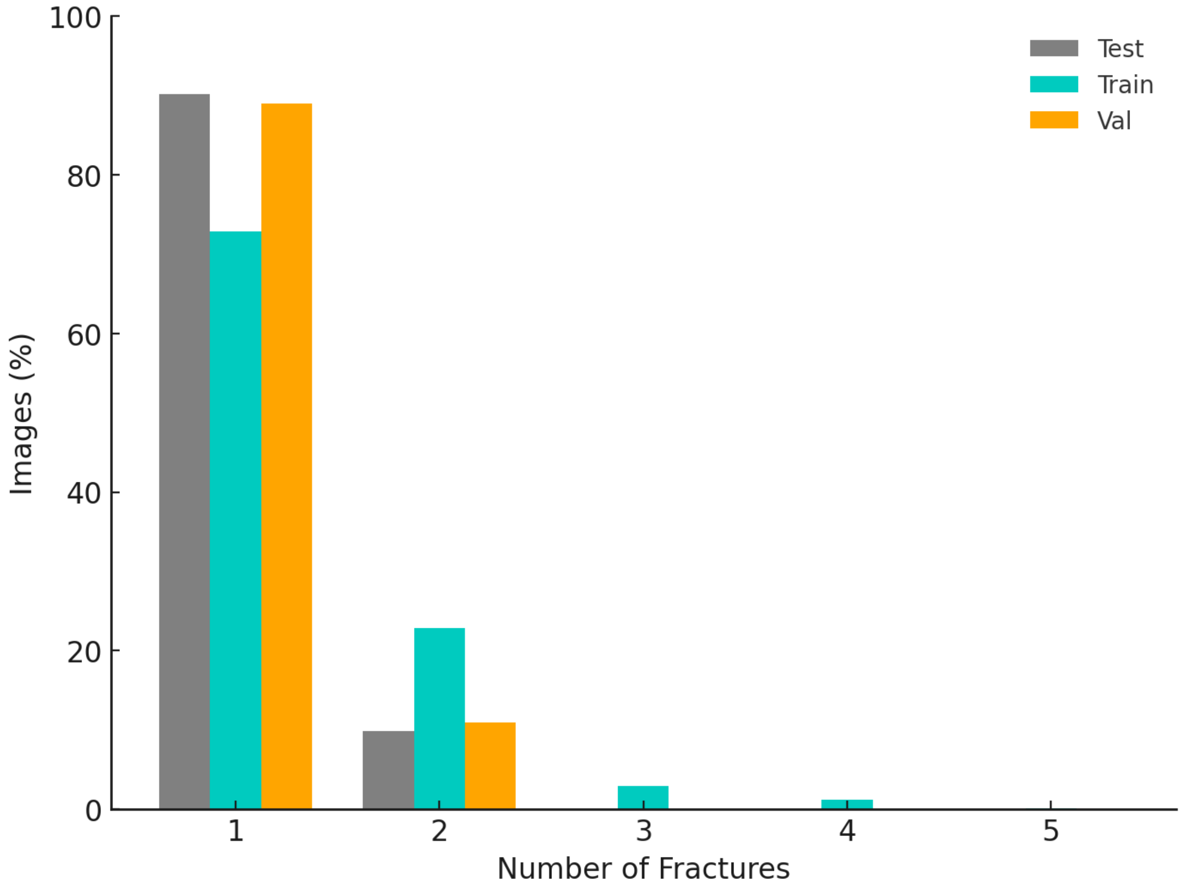
Dataset



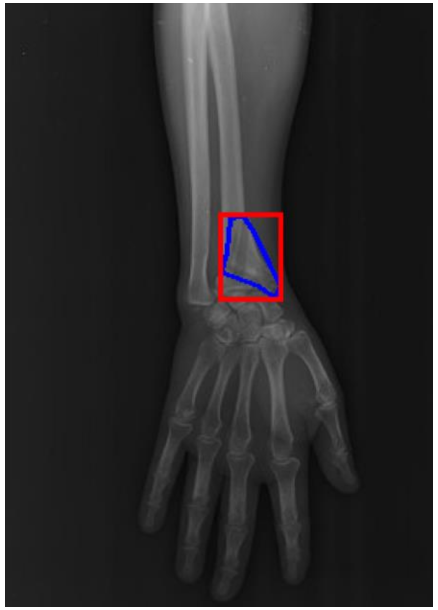

test
train
val
Number of Images
0
5
10
15
20
25
30
35
50
100
150
200
250
300
350
40
hand
hip
leg
shoulder


test
train
val
Number of Images
0
5
10
15
20
25
30
50
100
150
200
250
40
frontal
lateral
oblique

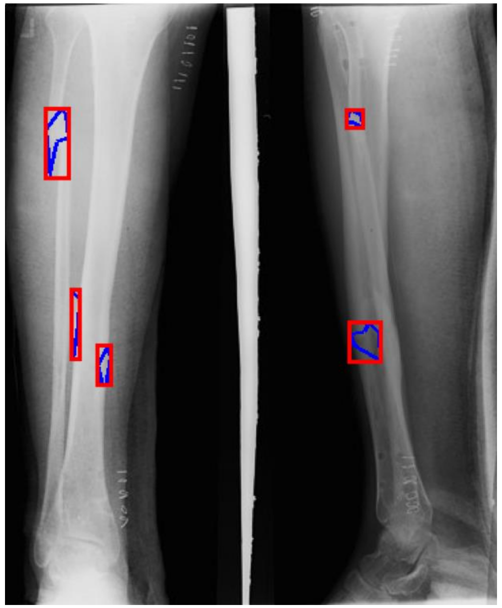
Dataset



1 fracture



5 fractures



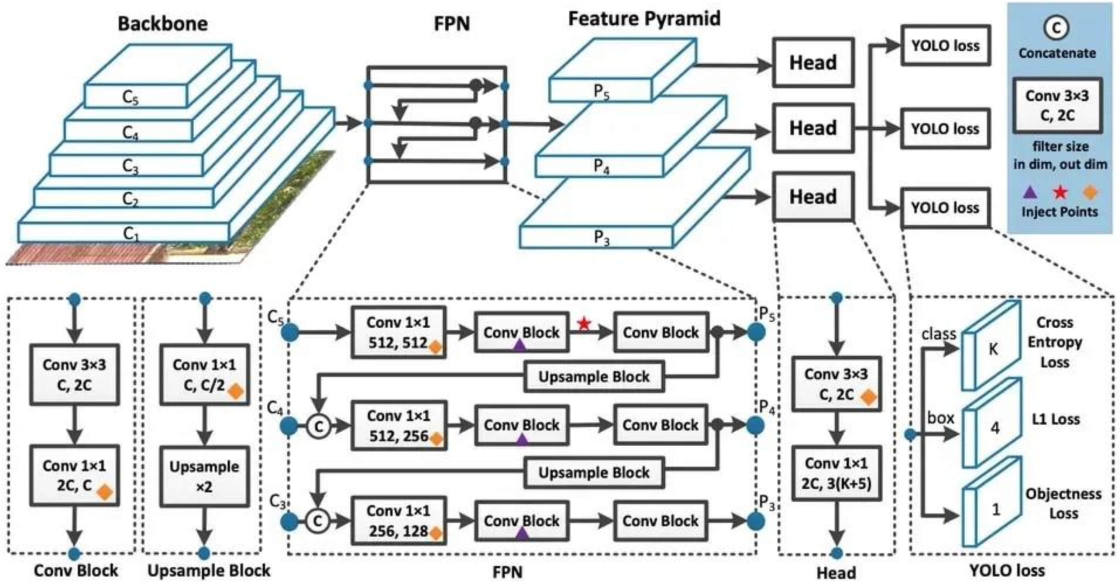
Methods

Object detection and Segmentation

- Yolo v8[2] and Yolo v11 models

Image Classification – "Benchmark"

- Vision Transformer[3] from scratch
- Transfer learning (only top trained) using ResNet50[4], VGG19[5] and Inception V3[6]



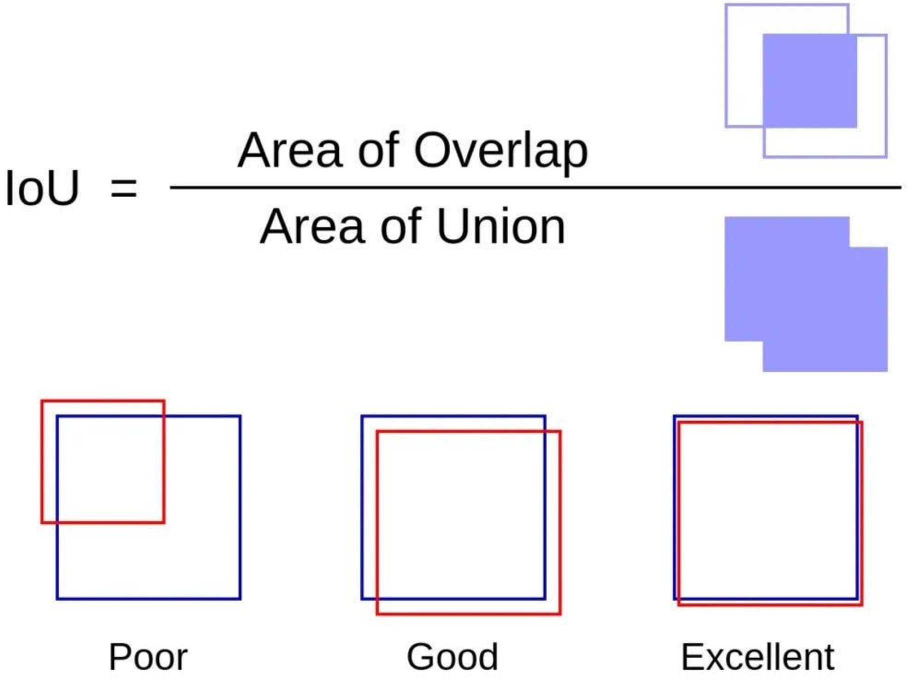
YOLO v8 – architecture

[2] Sohan et al. (2024). A review on yolov8 and its advancements. In *International Conference on Data Intelligence and Cognitive Informatics* (pp. 529-545). Springer, Singapore.
[3] Dosovitskiy et al. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929.
[4] He et al. (2016). Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 770-778).
[5] Simonyan & Zisserman (2014). Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556.
[6] Szegedy et al. (2016). Rethinking the inception architecture for computer vision. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 2818-2826).

Methods

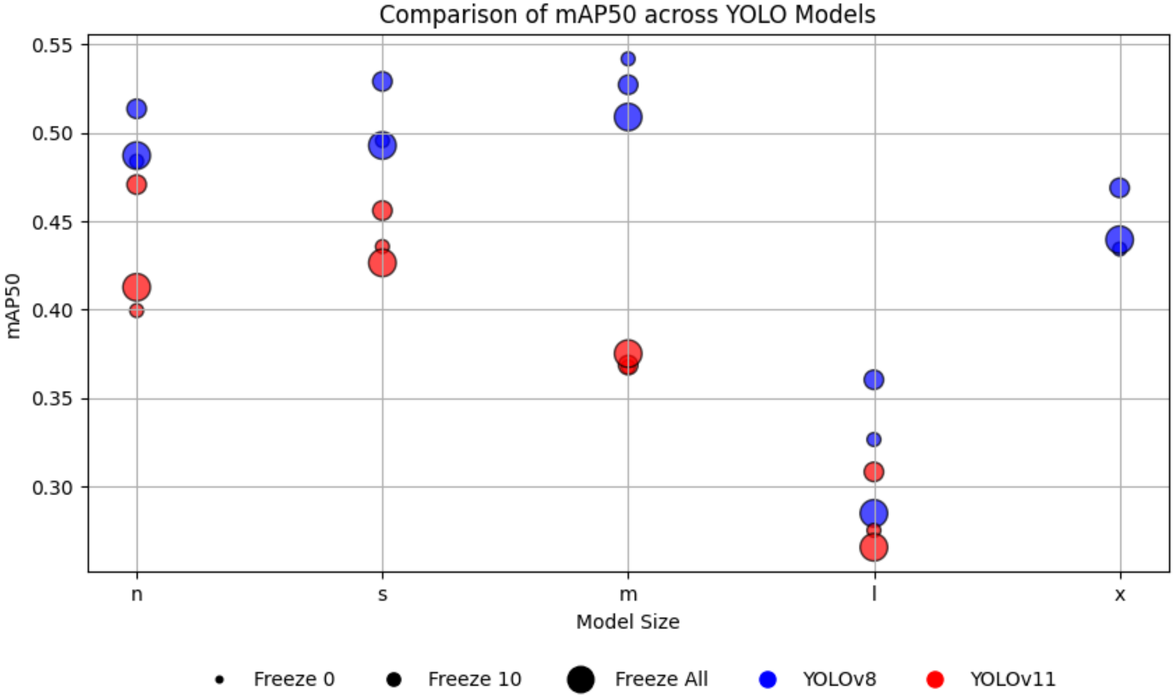
Metrics

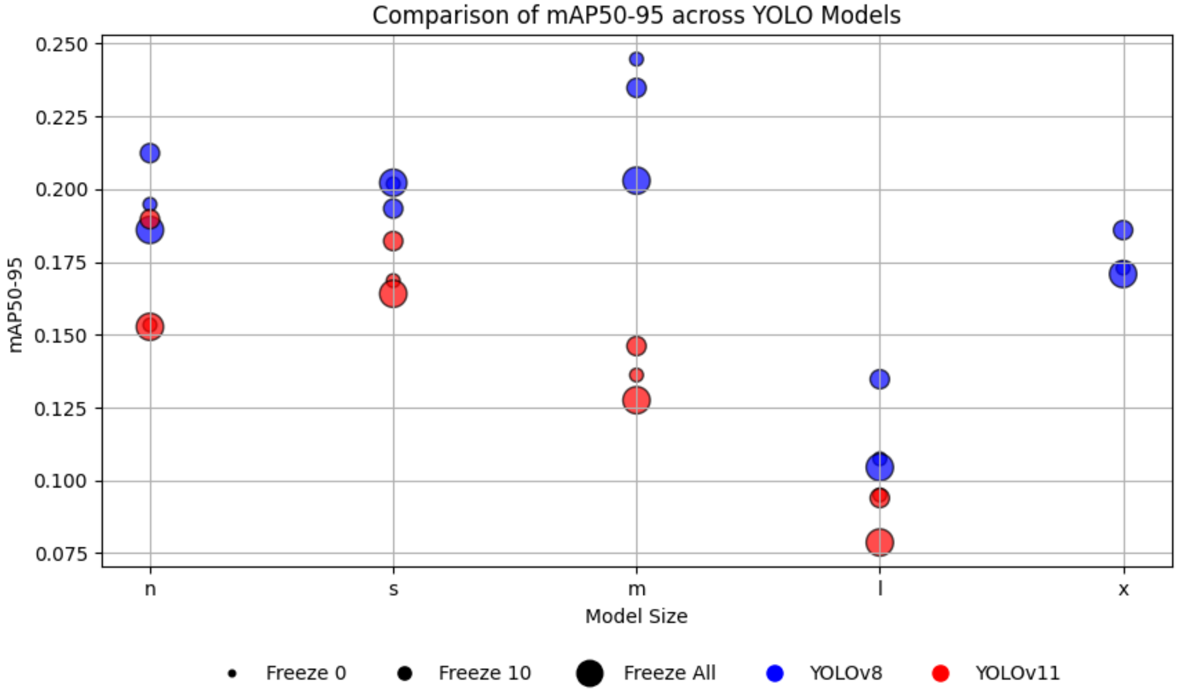
- Mean Average Precision
  - mAP50: @ intersection over union (IoU) threshold of 0.50
  - mAP50-95: @ varying IoU thresholds, ranging from 0.50 to 0.95
- Accuracy
- Precision
- Recall
- F1-Score



https://nikhilroxtomar.medium.com/what-is-intersection-over-union-iou-in-object-detection-idiot-developer-39448ac56431

Object detection - YOLO

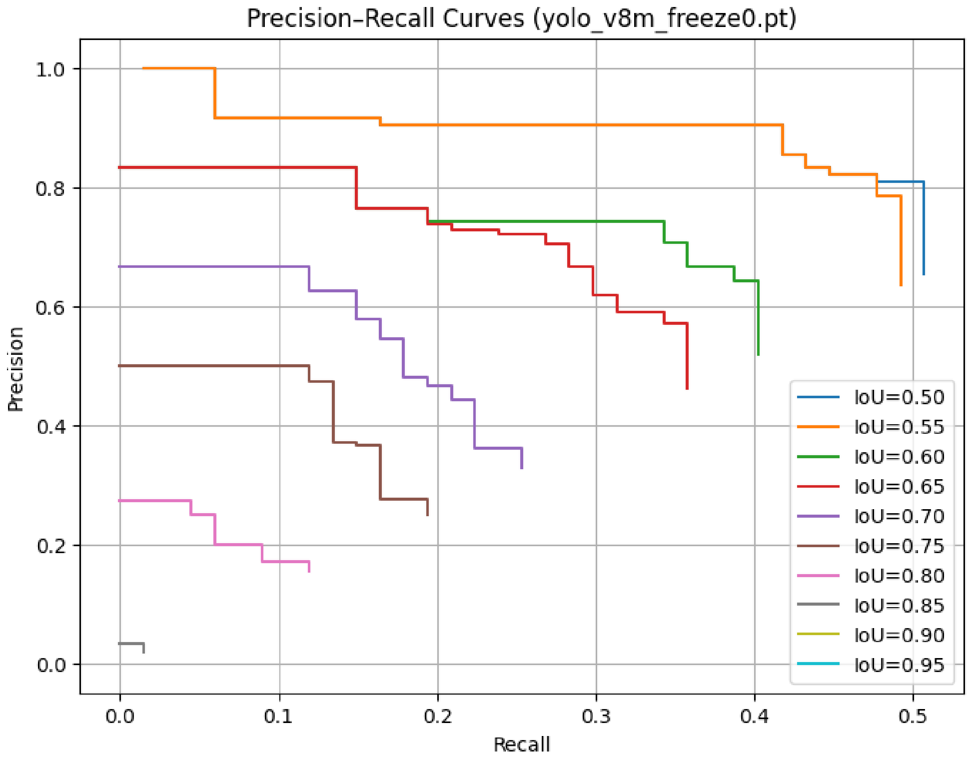


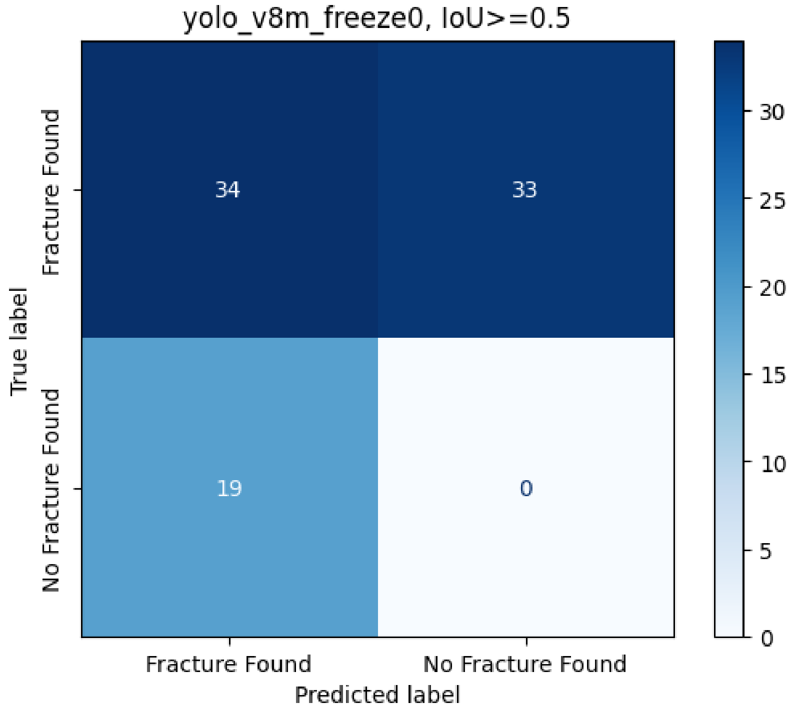


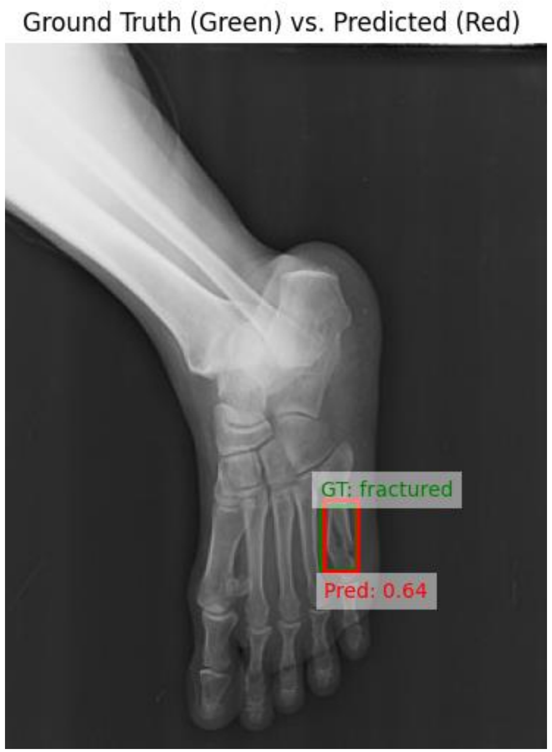
using only images containing ‘fractured’ objects

Results – unseen test data

YOLO v8m (0 freeze layers)

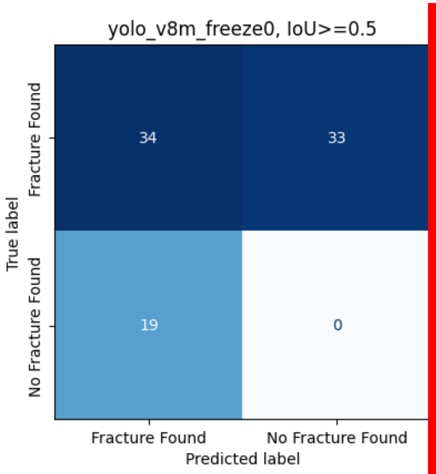




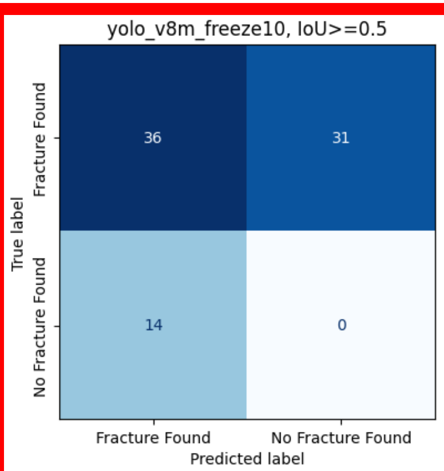


Precision: 0.64 | Recall: 0.51 | Specificity: 0.00 | F1-score: 0.57

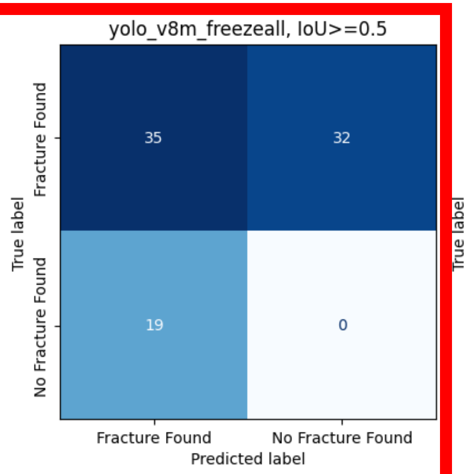
Results – Top 5



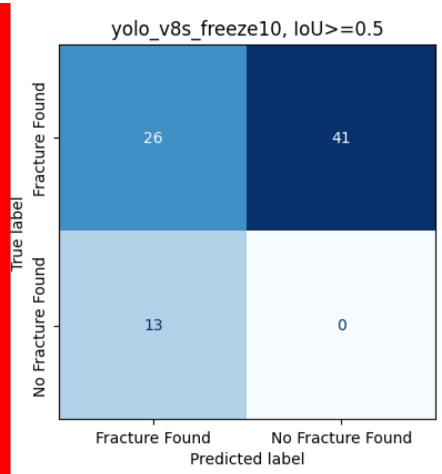
Precision: 0.64
Recall: 0.51
Specificity: 0.00
F1-score: 0.57



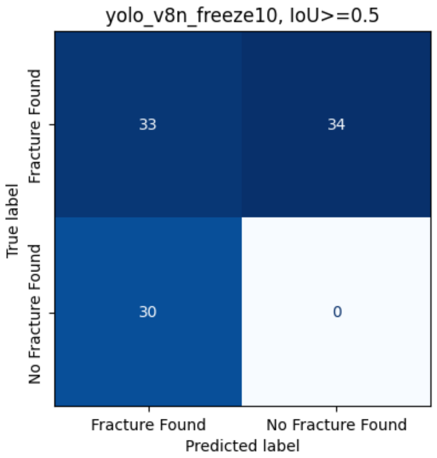
Precision: 0.72
Recall: 0.54
Specificity: 0.00
F1-score: 0.62



Precision: 0.65
Recall: 0.52
Specificity: 0.00
F1-score: 0.58

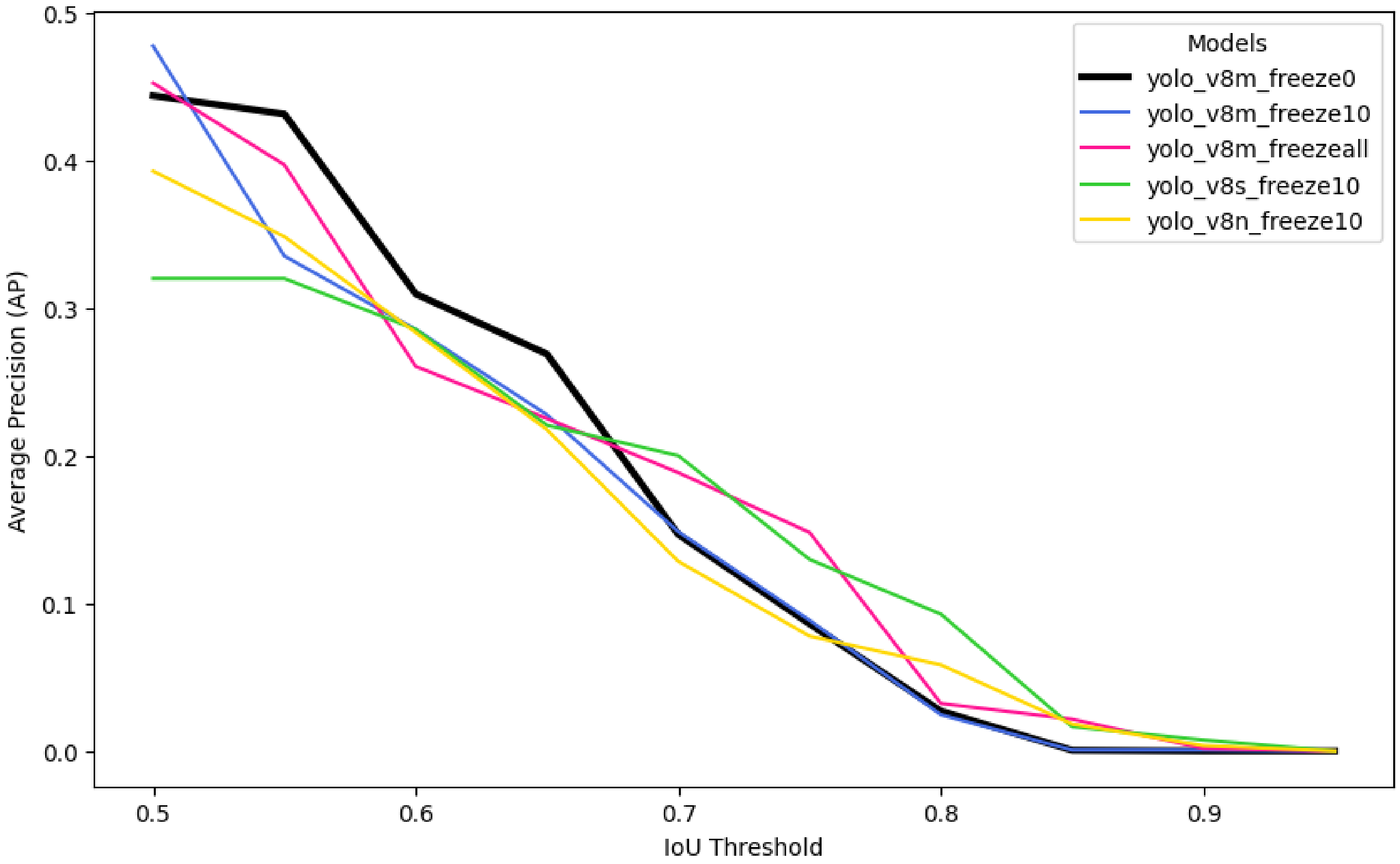


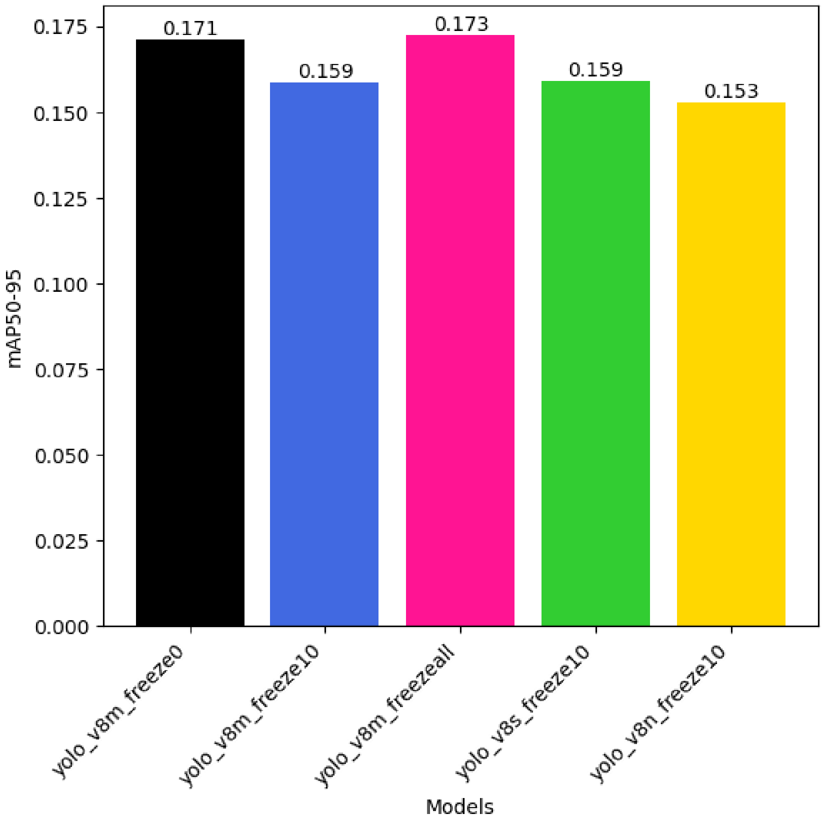
Precision: 0.67
Recall: 0.39
Specificity: 0.00
F1-score: 0.49



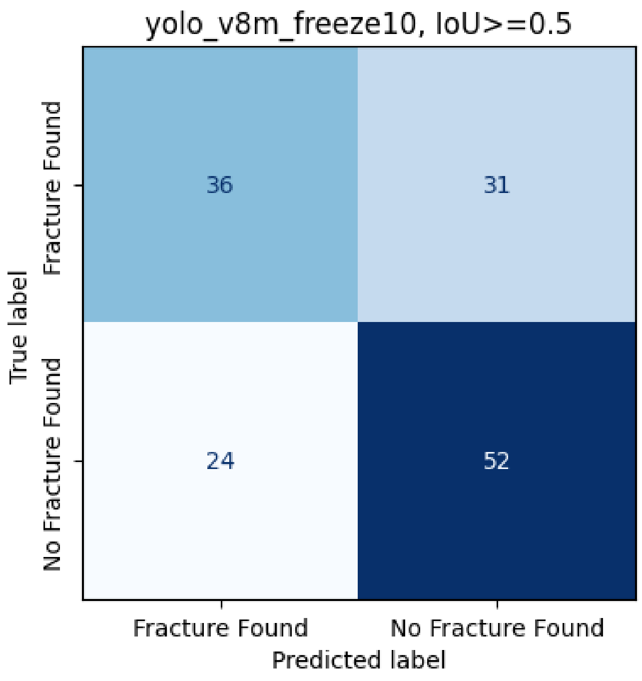
Precision: 0.52
Recall: 0.49
Specificity: 0.00
F1-score: 0.51

Results – Top 5

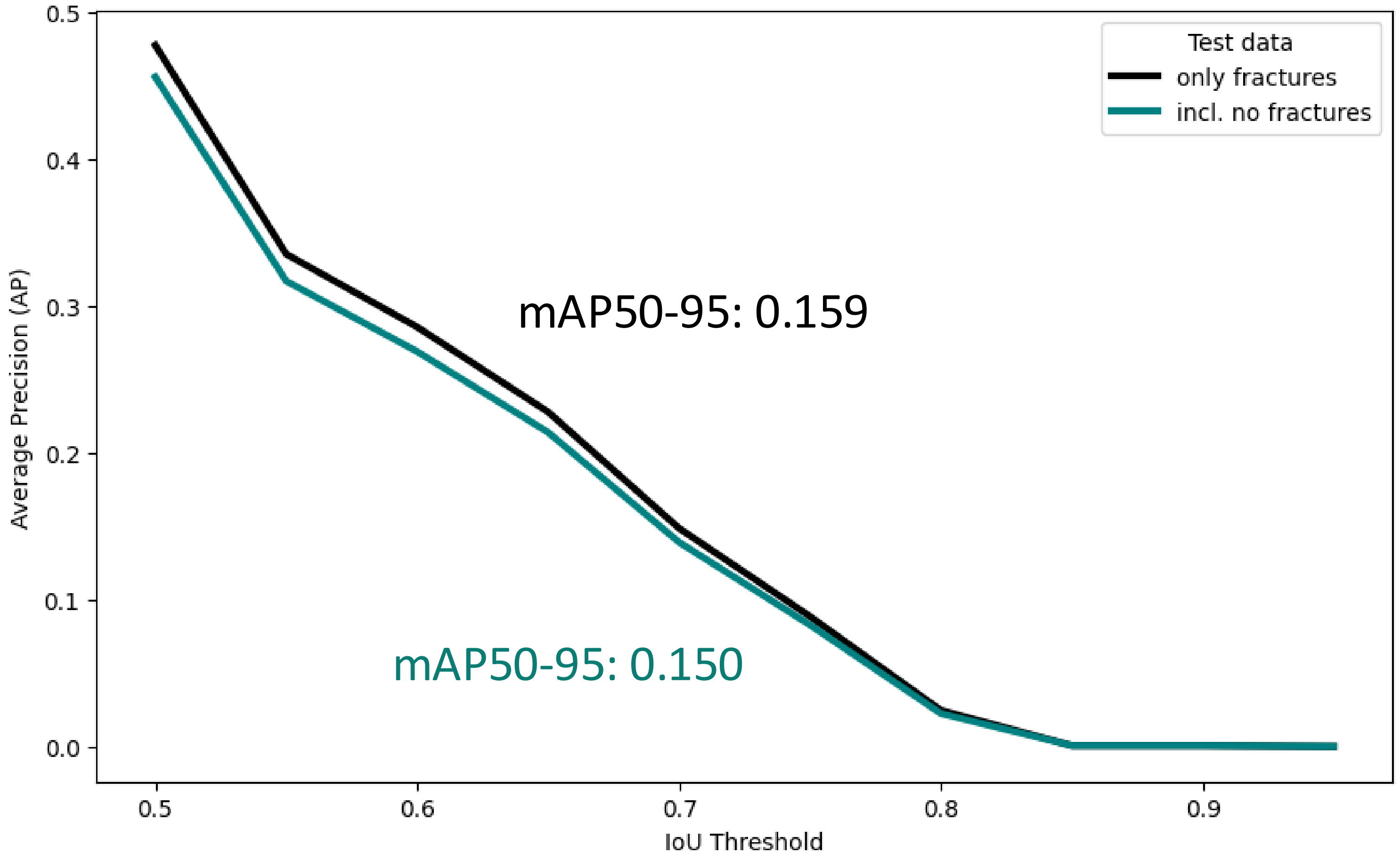




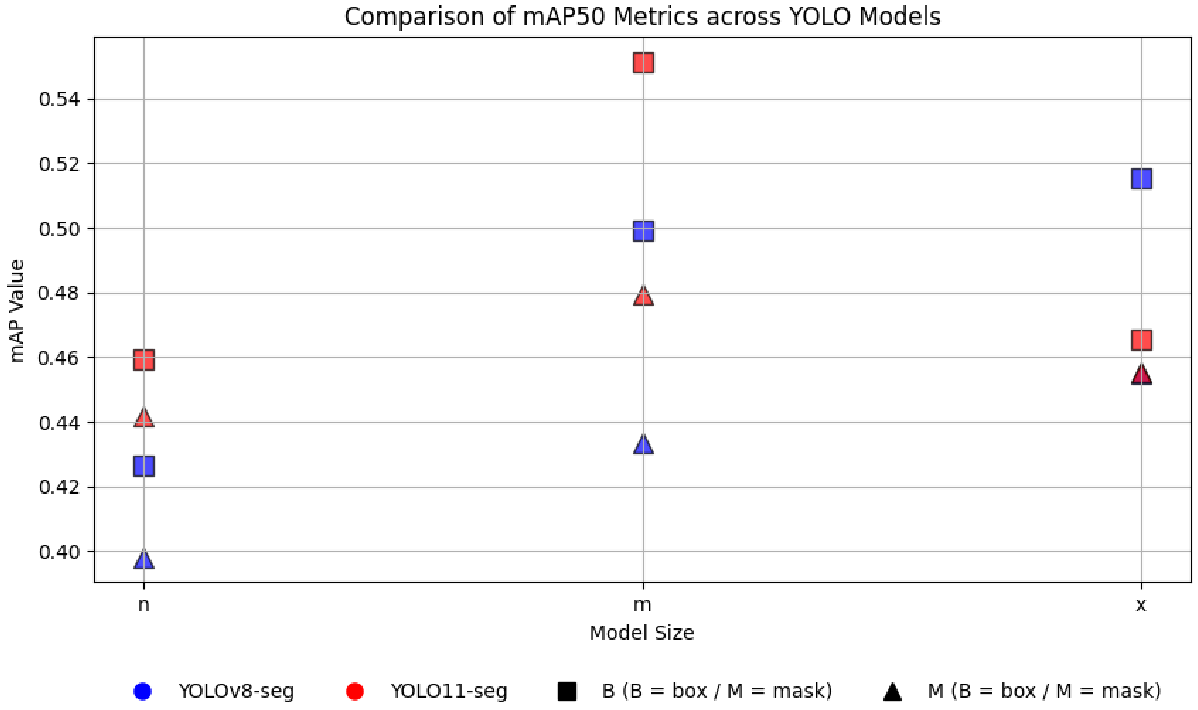
Results – test data incl. "no fractures"

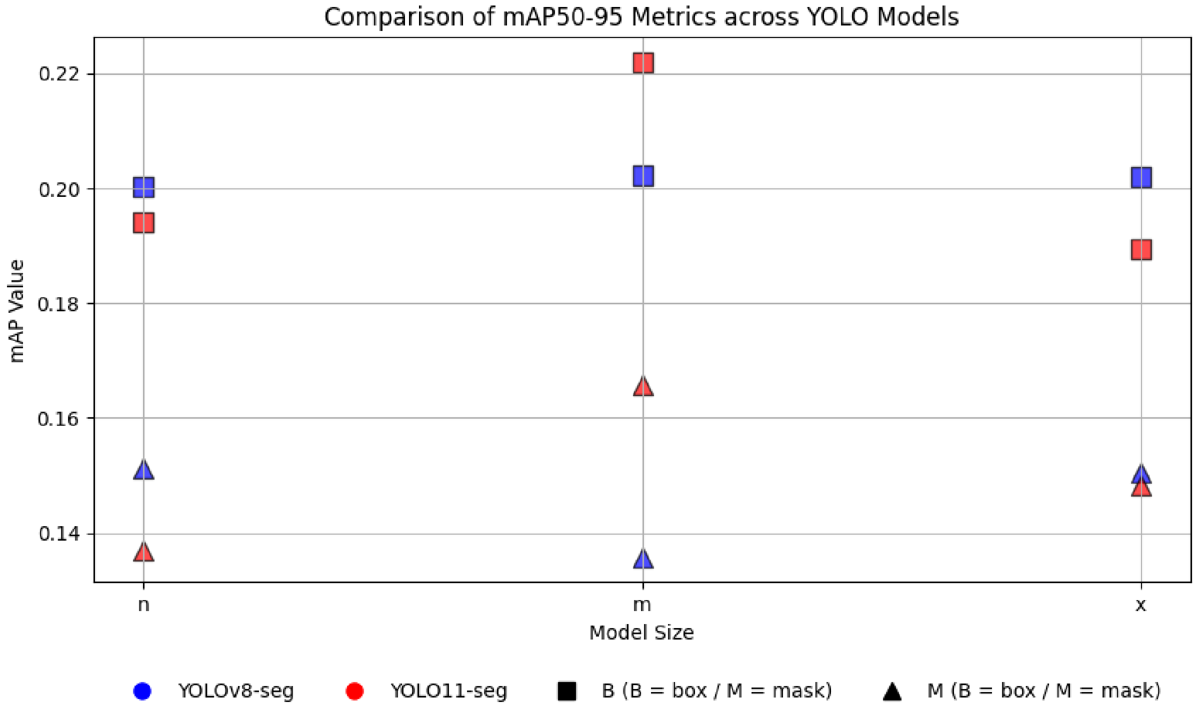


Precision: 0.60
Recall: 0.54
Specificity: 0.68
F1-score: 0.57



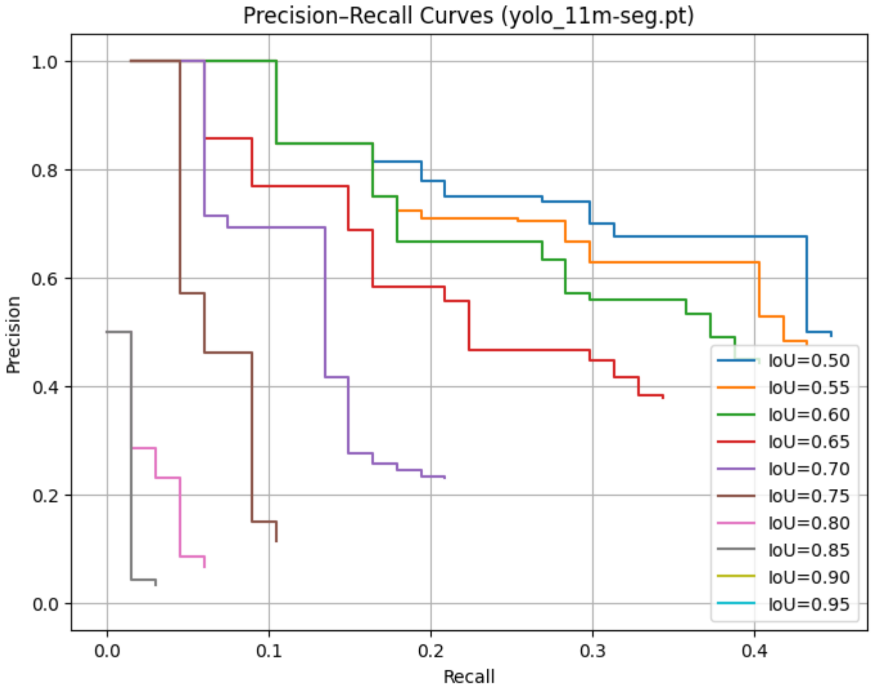
Instance Segmentation - YOLO

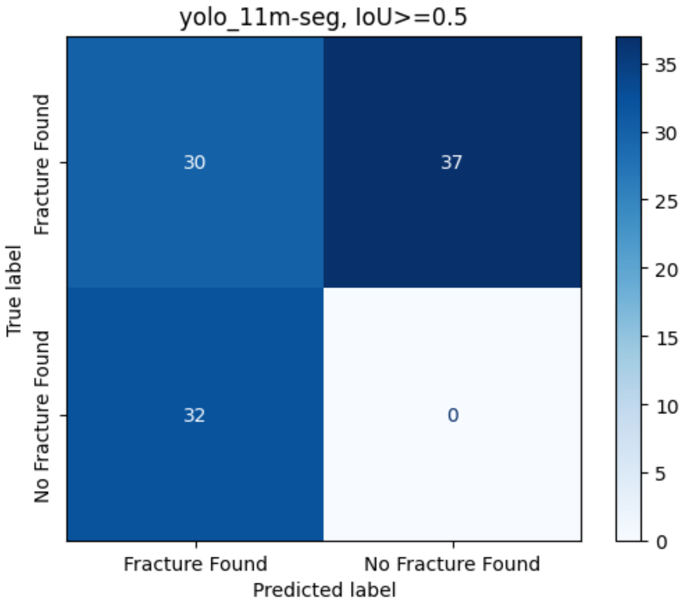




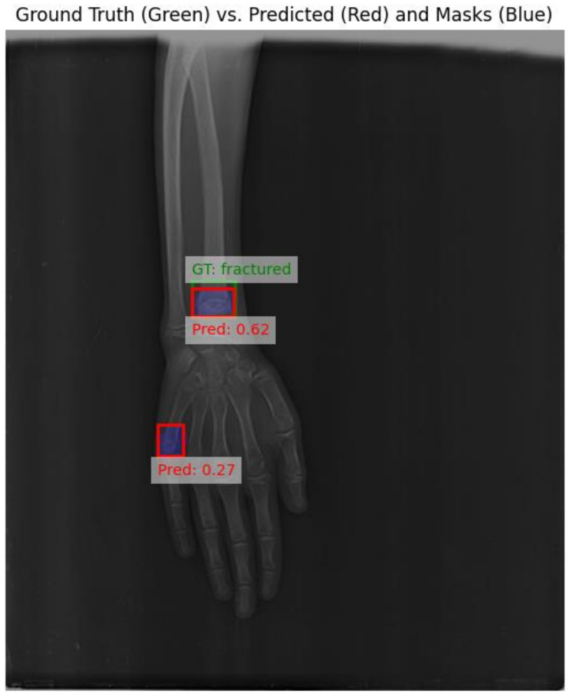
Results segmentation – unseen test data

YOLO11m-seg (10 freeze layers)





Precision: 0.48 | Recall: 0.45 | Specificity: 0.00 | F1-score: 0.47



Model comparison

Object Detection

	YOLO v8 (backbone)	YOLO v8 (incl. not fractured)	Segmentation
mAP50	0.48	0.46	0.34
mAP50-95	0.16	0.15	0.14

Model comparison

Object Detection

	YOLO v8 (backbone)	YOLO v8 (incl. not fractured)	Segmentation
mAP50	0.48	0.46	0.34
mAP50-95	0.16	0.15	0.14
Accuracy	0.44	0.62	0.30
Precision	0.72	0.60	0.48
Recall	0.54	0.54	0.45
F1-Score	0.62	0.57	0.47

Model comparison

	Object Detection				Transfer Learning		
	YOLO v8 (backbone)	YOLO v8 (incl. not fractured)	Segmentation	Vision Transformer	ResNet50	VGG19	Inception V3
mAP50	0.48	0.46	0.34				
mAP50-95	0.16	0.15	0.14				
Accuracy	0.44	0.62	0.30	0.57	0.69	0.72	0.70
Precision	0.72	0.60	0.48	0.60	0.68	0.70	0.72
Recall	0.54	0.54	0.45	0.43	0.70	0.77	0.67
F1-Score	0.62	0.57	0.47	0.50	0.69	0.73	0.69

Discussion

Chen et al, (2024)[7] - Weighted-channel-attention-YOLO v8:

- **FracAtlas**:
  Baseline (YOLOv8-s): mAP50: 0.48, mAP50-95: 0.18, precision 0.80, recall 0.41
  WCA & DSC-Cf2: mAP50: 0.57, mAP50-95: 0.23, precision 0.76, recall 0.52
- **GRAZPEDWRI-DX** (pediatric wrist trauma dataset)[8] : mAP50: 0.64, detection accuracy 0.94

	YOLO v8 (backbone)
mAP50	0.48
mAP50-95	0.16
Accuracy	0.44
Precision	0.72
Recall	0.54
F1-Score	0.62

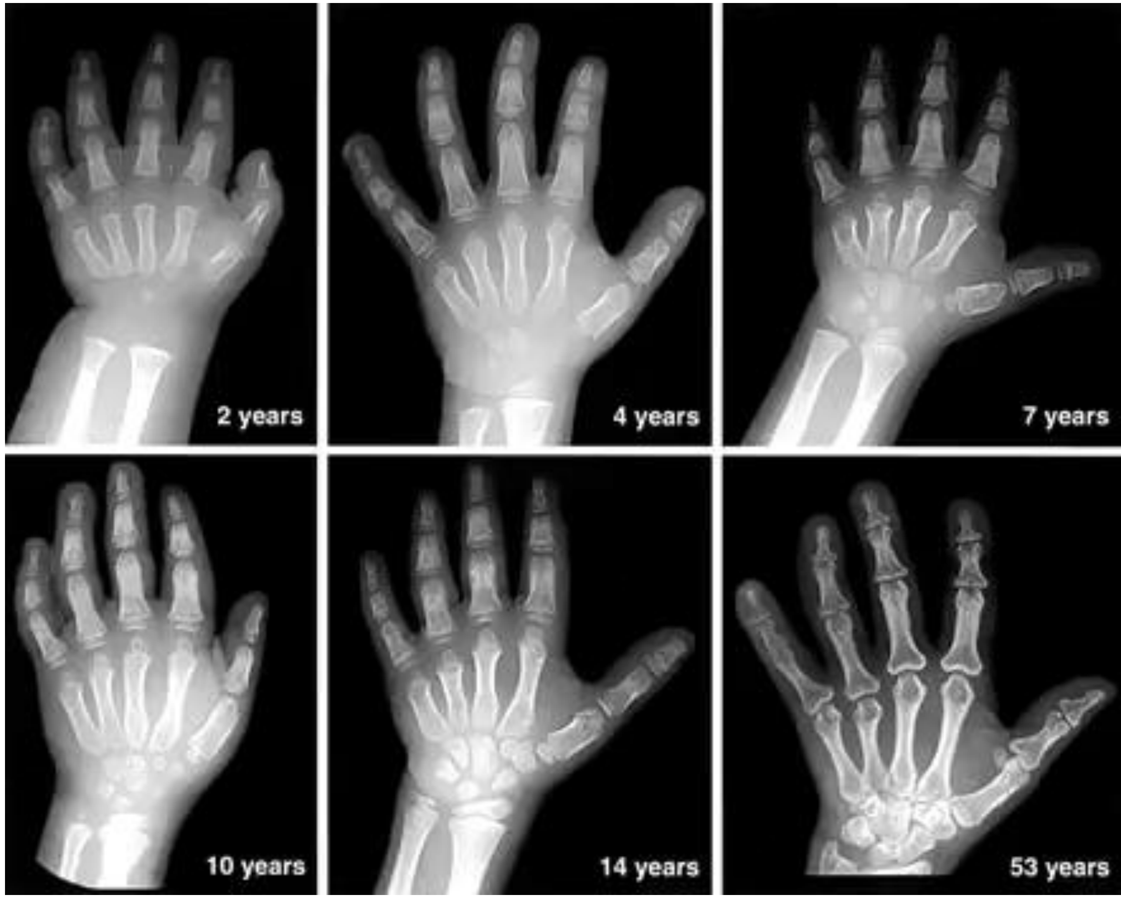
- YOLOv8 tuned for GRAZPEDWRI-DX[9]: mAP50 = 0.64
- ParallelNet[10] on Thigh Fracture Dataset[11]

Method	Backbone	AP50 (%)
DCFPN	ResNet-101	82.1
Faster R-CNN	ResNet-50	78.1
FPN	ResNet-101	85.2
Cascade R-CNN	ResNet-50	85.0
FPN (guided anchoring)	ResNet-101	85.1
RetinaNet (with FPN)	ResNet-101	85.3
ParallelNet (ours)	TripleNet (ours)	87.8

[7] Chen, et al. (2024). WCAY object detection of fractures for X-ray images of multiple sites. Scientific Reports, 14(1), 26702.
[8] Nagy et al. (2022) A pediatric wrist trauma X-ray dataset (GRAZPEDWRI-DX) for machine learning. *Sci Data.Bold, 9*, 222.
[9] Ju & Cai (2023) Fracture detection in pediatric wrist trauma X-ray images using YOLOv8 algorithm. *Sci Rep* **13**, 20077.
[10] Wang et al. (2021) ParallelNet: Multiple backbone network for detection tasks on thigh bone fracture. Multimedia Systems, 1-10.
[11] Guan et al. (2019) Thigh fracture detection using deep learning method based on new dilated convolutional feature pyramid network. Pattern Recognition Letters, 125, 521-526.

Limitations

- Limited number of available samples; and those are at several different locations
- X-rays from children and adults
  → different status of ossification

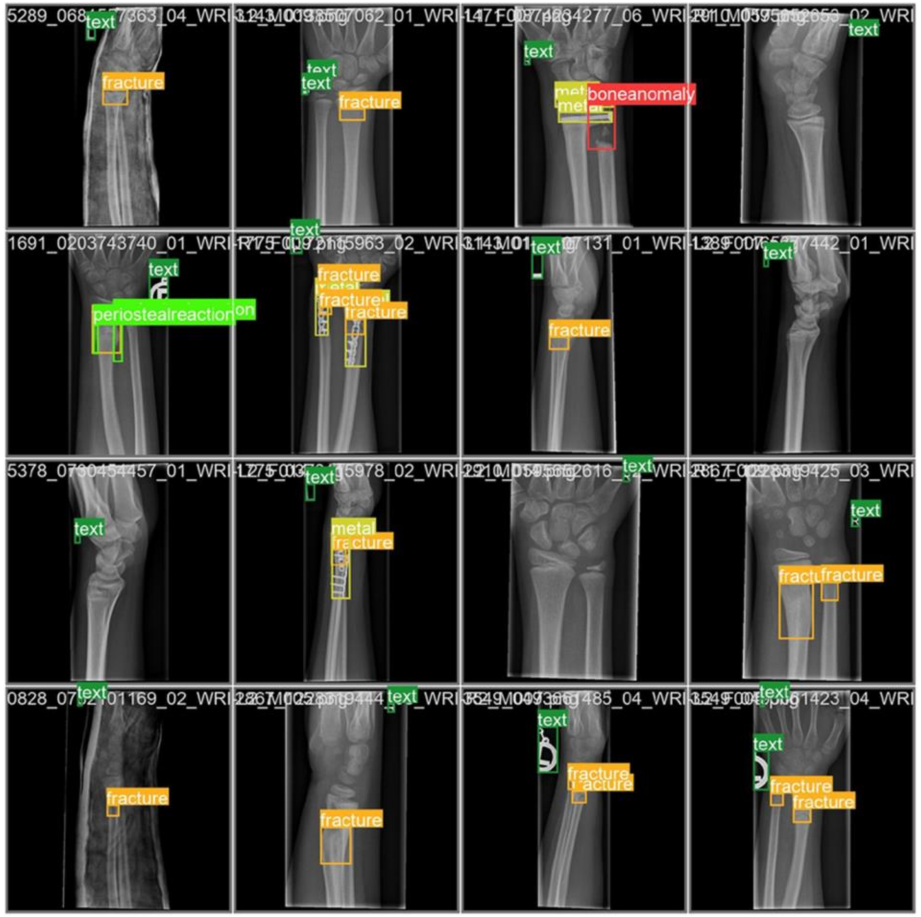


Limitations

- Limited number of available samples; and those are at several different locations

- X-rays from children and adults
  → different status of ossification

- Other x-ray datasets with more object annotations[8,9]



[8] Nagy et al. (2022) A pediatric wrist trauma X-ray dataset (GRAZPEDWRI-DX) for machine learning. *Sci Data.Bold, 9*, 222.
[9] Ju, RY., Cai, W. (2023) Fracture detection in pediatric wrist trauma X-ray images using YOLOv8 algorithm. *Sci Rep* **13**, 20077.

Future directions

- Dataset Improvements
  - o Increase the number of X-ray images for better generalization using a automatic segmentation tool (see next slide)
  - o Balance the dataset across different age groups and body regions
- Model Training Enhancements
  - o Train models separately for different body regions to improve accuracy

Individual segmentation tool

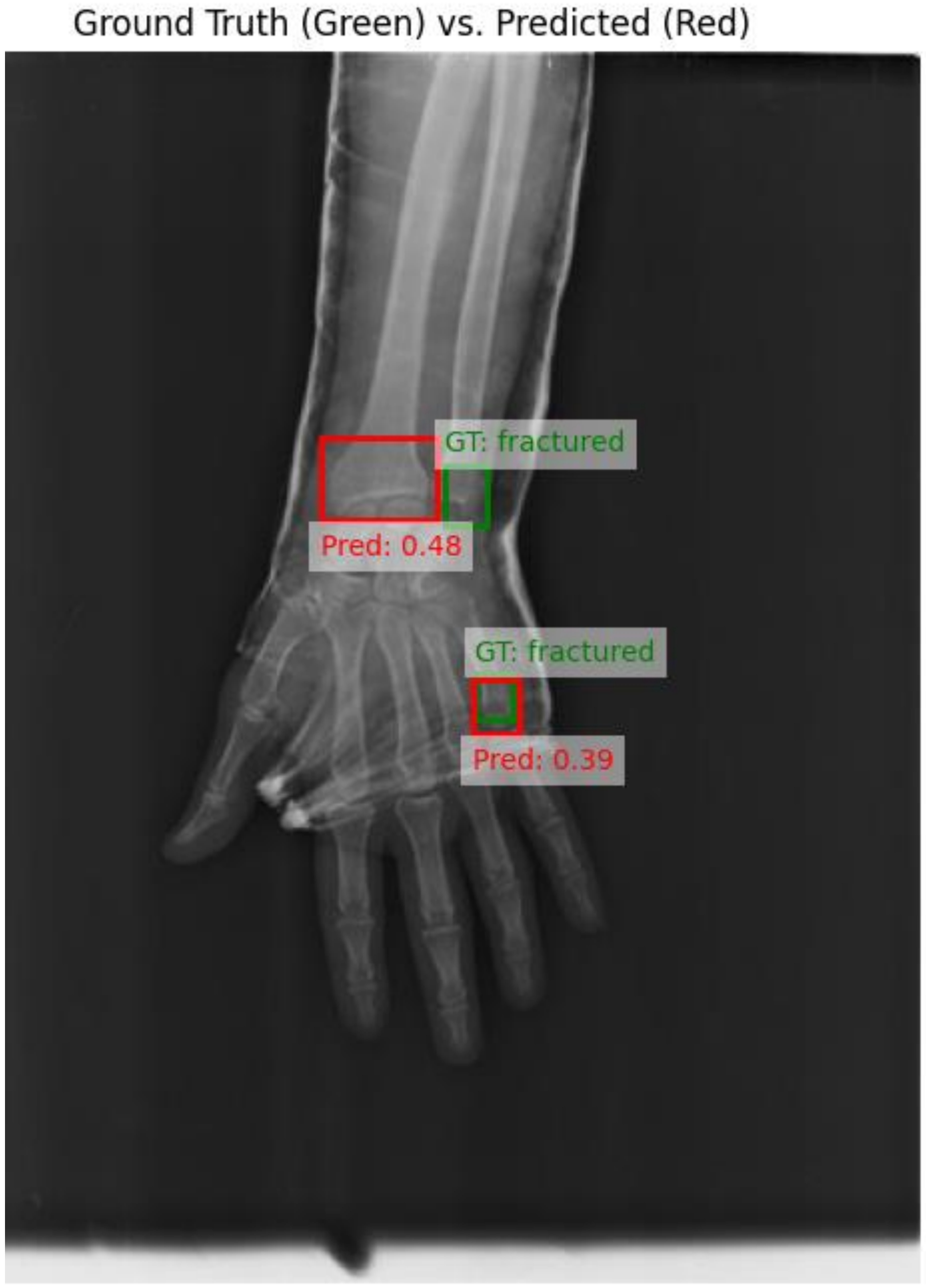
File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python 3 (ipykernel)

```
        "segmentation": segmentation,                              # [x1, y1, x2, y2, ..., xn, yn]
        "area": area,
        "bbox": bbox,
        "iscrowd": 0                          # crowded object: where boundaries are not well defined due to overlapping/occlusion
    }
    return annotation

def predict_images(image, input_points, input_labels, image_path, image_id, image_folder):
    predictor.set_image(image)
    input_points = []
    input_labels = []
    set_first_points(image, input_points, input_labels, image_path, image_id, image_folder)
```

[6]:
```
# User selects the folder
image_folder = select_folder()
print(f"Selected folder: {image_folder}")

# Get the first image in the folder
if image_folder:
    image_files = [f for f in os.listdir(image_folder) if f.lower().endswith(('.png', '.jpg', '.jpeg'))]
    print(f"Total {len(image_files)} images found in selected folder")

    if image_files:
        first_image_path = os.path.join(image_folder, image_files[0])
        image_path = first_image_path
        image_id = 1
        print(f"Opening first image: {first_image_path}")

        # Load the image
        image = Image.open(first_image_path)
        image = np.array(image.convert("RGB"))

        # Display the image
        plt.imshow(image)
        plt.axis("on")
        plt.show()
    else:
        print("No images found in the selected folder.")
else:
```

Bone fracture detection from X-rays



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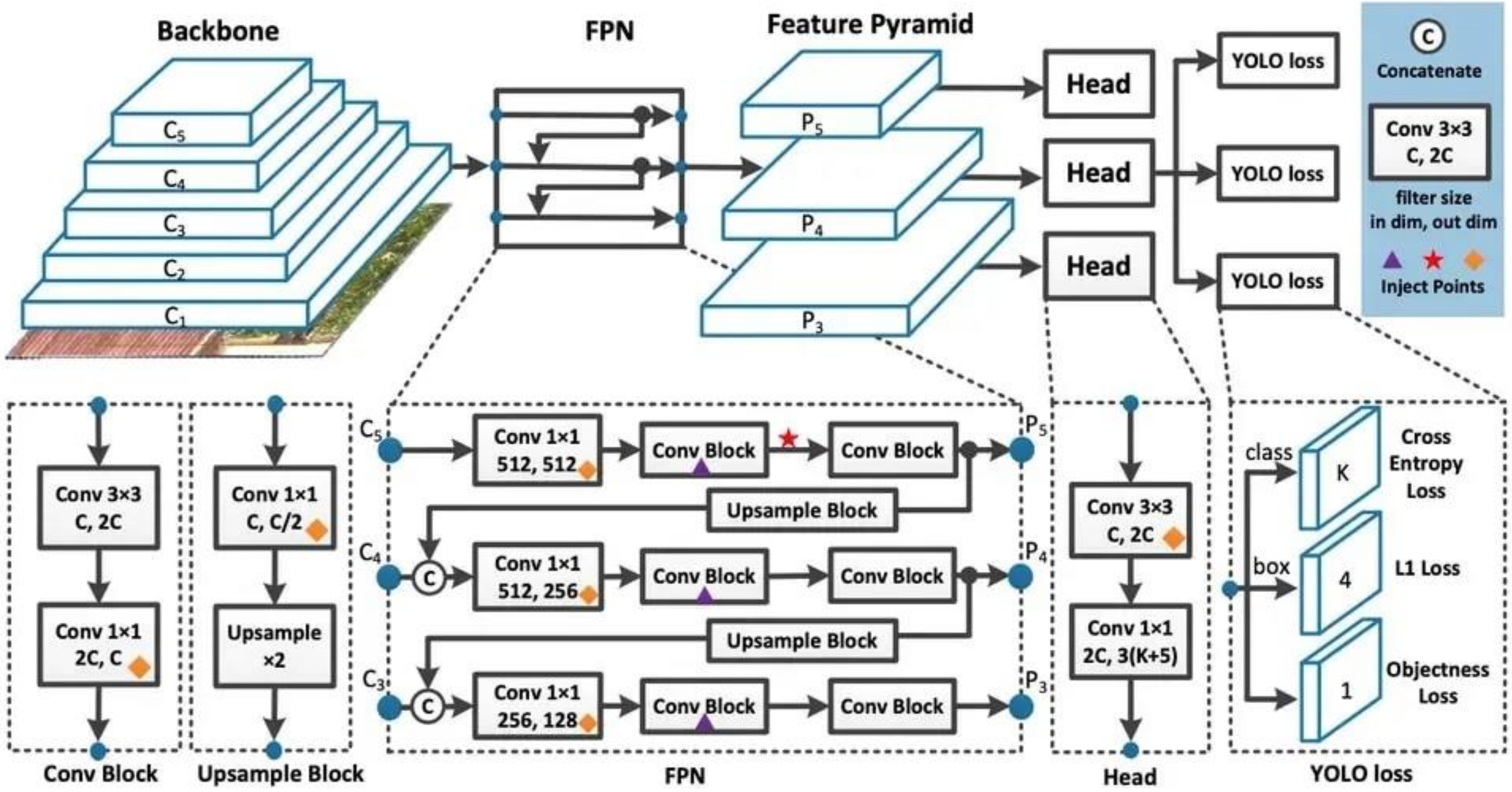
YOLO v8



Backbone
FPN
Feature Pyramid
C_5
C_4
C_3
C_2
C_1
P_5
P_4
P_3
Head
Head
Head
YOLO loss
YOLO loss
YOLO loss
Concatenate
Conv 3×3 C, 2C
filter size in dim, out dim
Inject Points
Conv 3×3 C, 2C
Conv 1×1 2C, C
Conv Block
Conv 1×1 C, C/2
Upsample ×2
Upsample Block
Conv 1×1 512, 512
Conv Block
Conv Block
Upsample Block
Conv 1×1 512, 256
Conv Block
Conv Block
Upsample Block
Conv 1×1 256, 128
Conv Block
Conv Block
FPN
Conv 3×3 C, 2C
Conv 1×1 2C, 3(K+5)
Head
class
box
K
4
1
Cross Entropy Loss
L1 Loss
Objectness Loss
YOLO loss